The Joy of Computing Using Python - Exam Practice Questions

Top 20 High-Probability Exam Questions

1. What is the formula for a magic constant of size n magic square?
2. What are the correct ways to concatenate strings in Python?
3. Difference between find() and index()?
4. Explain the purpose of a base case in recursion.
5. How does a binary search differ from a linear search in performance?
6. Output of: dict_name = {1: 'one', 1.0: 'float one', True: 'boolean one'}
7. Which code block prints: "Ramesh talks to Suresh and Kamlesh"?
8. Output of: tup = ([1, 2, 5], [3, 4]); tup[0].append(5)
9. Purpose of penup() in turtle graphics?
10. Which turtle command draws a hexagon?
11. How many distinct winning lines pass through center of Tic-Tac-Toe board?
12. How does PageRank determine importance of a web page?
13. What does courses['CS102'] = 'DBMS' do in a dictionary?
14. How can frequency analysis help in breaking substitution cipher?
15. What does thumbnail() do in PIL?
16. What does mystery(n) return if defined as: return n + mystery(n - 1)?
17. How many swaps in third iteration of bubble sort on [5, 4, 3, 2, 1]?
18. Output of print(17 % -5)?
19. What does G.degree(node) return in NetworkX?
20. Identify Social Computing examples (e.g., Wikipedia, ChatGPT, Twitch, etc.)